Fiji Meteorological Service

Software Requirements Specification Document

For Fiji MET:

Flood Risk Prediction and Community Alert System

Version 1.0
Prepared by Group 13
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Revision History

Name	Date	Reasons for Change	Version
Initial	28/08/2026	-	1

1.0 Introduction

- 1.1 Purpose

The software requirements specification (SRS) identifies the requirements of the Flood Risk Prediction and Community Alert System. This document will outline the overall description of the system, functional and non-functional requirements of the system. This will allow internal and external users to better understand the system prior to release to ensure development satisfies user's known and unknown needs.

- 1.2 Document Conventions

The SRS is created in a manner that is easy to follow, easy to find specific topics and make it easier to understand what the end product will roughly function. Using numbered topics (1.0, 2.0, etc) and sub-topics that are ordered from most brief to most detailed. Moreover, certain features may or may not be implemented depending on feedback from the client.

- 1.3 Target Audience and Reading Suggestions

The SRS is intended to better educate internal and external stakeholders.

1. Development Team
   It is advised for the development team to read through:
   - 3.0
   - 4.0
   - 5.0
2. Project Manager/Scrum Master
   It is advised for the PM and SM to read through:
   - 2.0
   - 3.0
   - 4.0
   - 5.0
3. Fiji Metrological Services
   It is advised for the client to read through:
   - 2.0

- 3.0
- 4.0
- 5.0

- 1.4 Product Scope

The Flood Risk Prediction and Community Alert System aims to enhance the Fiji Metrological Service's web-based platform using artificial intelligence to predict and alert flood risks for different areas in Fiji. The system is planned to display predictions via a simple and non-technical dashboard and weather map that the general public can understand. It further aims to assist Fiji Metrological Services to replace manual updates with automatic alerts for faster, more informed and clear data-driven updates as voiced out by the client (Fiji Times, 2026).

- 1.5 References

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2.0 Overall Description

2.1 Product Perspective

The Flood Risk Prediction and Community Alert System is a new web-based system designed to support the Fiji Meteorological Service in monitoring and communicating flood risk across Fiji. The system aims to improve the existing process of analysing weather information and communicating flood warnings by incorporating artificial intelligence to generate location-based flood-risk predictions.

The system will operate as a web application accessible through modern desktop and mobile browsers. The frontend will communicate with FastAPI backend services through REST API endpoints, while the backend will manage communication between the database, weather data sources, AI prediction model and notification services.

2.2 Product Functions

The main functions of the Flood Risk Prediction and Community Alert System are divided between the community/public interface and the Fiji MET administrative interface. The major product functions are:

1. **User Registration and Authentication** - Registered users will be able to create accounts and securely log in to access personalized functions such as saved locations and notification preferences. Fiji MET administrators will have privileged accounts for accessing administrative functions.
2. **Weather Data Management** - The system will obtain weather and environmental data, validate the received information and store valid records for use by the flood-risk prediction model.
3. **AI Flood Risk Prediction** - The system will use an AI/ML model to analyse available weather and environmental data and generate flood-risk predictions for monitored locations. Predictions will be categorized as Low, Moderate, High or Very High risk.
4. **Flood Risk Dashboard** - Users will be able to view current flood-risk information, predictions, monitored or saved locations and relevant alerts through a simple dashboard.
5. **Flood Risk Map** - The system will provide an interactive map of Fiji displaying monitored locations and colour-coded flood-risk levels. Users will be able to select locations to view more detailed risk information.
6. **Alert Management** - Authorized Fiji MET administrators will be able to create, review, approve, override and suppress flood alerts. The AI model may generate candidate alerts when High or Very High flood risk is predicted, but an administrator must approve an alert before it is issued publicly.
7. **Alert Notifications** - Approved alerts will be displayed to the public through the dashboard and flood-risk map. Registered users who opt in to notifications will also be able to receive notifications for the following locations.
8. **Monitored Location Management** - Fiji MET administrators will be able to add, edit and remove locations monitored by the system.
9. **Reporting and Analytics** - Fiji MET administrators will be able to view system metrics and generate reports using stored flood-risk predictions, monitored-location information and alert data.

2.3 Operating Environment

The system will be developed in a web based environment. Here, the Fiji MET administrators add/edit information on flood risk and locations which are currently being monitored by them. Members of the public can view the information on flood risk, can view the maps and receive alerts by using their web browser to access this system to view the information.

2.3.1 Technical Environment

- The application will be developed as a web-based application. Therefore, it will be possible to view the application on desktop and mobile web browsers.
- The frontend will communicate with the backend through REST API endpoints.
- The frontend of the system will be a React-based application which will display the dashboard, the flood-risk map, the alerts and the rest of the views of the system.
- Backend services handle weather, monitored locations, flood risk prediction and alerting. It also deals with database communication.
- The system will use PostgreSQL/PostGIS for relational and geographic data storage.
- The AI/ML component for flood-risk prediction based on weather and environmental data will be integrated into the system.
- This application requires an active internet connection in order to obtain current weather data and return to the user current flood risk for areas of interest.
- The deployment of the application may use containerization with Docker to provide a consistent environment for the application to run in.

2.3.2 Hardware Environment

No special hardware is required for normal users to use the system as it will be mainly a web application.

- A wide variety of devices can be used to access the system including desktops, laptops, tablets and even smartphones.
- User devices will require a supported web browser and an Internet connection to operate.
- The server environment must have sufficient processing power in order to run the backend services, the database and the AI prediction functionality.
- The server must have sufficient storage to hold weather data, users locations, users information, system's predictions, alerts and system logs.
- There must be sufficient bandwidth to support all users accessing the Flood Risk Dashboard and flood-risk map at the same time.
- Administrator devices require an Internet connection to add/edit locations, edit threshold values, add/edit predictions and associated alerts.

2.3.3 Software Environment

The system will operate using the following software applications:

- Visual Studio Code (VS Code) – primary Integrated Development Environment (IDE) for the code of the development team for the whole project for writing, editing, debugging.
- React – frontend development and user-interface components.
- FastAPI – backend API development and communication between the frontend, database and AI services.
- Python – backend and machine-learning development.
- PostgreSQL – relational database management.
- PostGIS – The PostgreSQL extension for Geographic/Spatial information. This will be used to store information about monitored locations and for flood mapping.
- Docker – containerisation of system components for consistent development and deployment.
- Git/GitHub – source-code version control, branch management and team collaboration.
- Modern web browsers such as Google Chrome, Microsoft Edge and Firefox.

2.3.4 User Environment

The system is intended to support two main user environments: community/public users and Fiji MET administrators.

- Community users can use a web browser on desktops or mobiles to view flood-risk information such as predictions, risk maps and alerts for the public.
- Fiji MET administrators will use their Fiji MET administrator accounts to access the Fiji MET administrators interface where they can add locations monitored by the system, change thresholds for locations monitored by the system, view latest predictions for locations monitored by the system, add, edit and delete public alerts and view reports for system usage.

2.4 Design and Implementation Constraints

2.4.1 Standards Compliance

The Flood Risk Prediction and Community Alert System should comply with standards set by the Fiji Institute of Engineers (FIE) and relevant international software engineering practices, including those set by the Institute of Electrical and Electronic Engineers (IEEE). The source code must follow consistent coding conventions and security guidelines to uphold the integrity and maintainability of the system, given its role in delivering time-sensitive public safety information.

2.4.2 Data Protection Compliance

The development team must ensure that user data and location-based flood risk data are protected from unauthorized access. The system shall implement access control mechanisms to restrict sensitive functionality such as alert approval, threshold configuration, and monitored location management to authenticated MET admin accounts only. The system should perform regular audits to review and detect unauthorized access attempts.

2.4.3 Hardware Limitations

The system must be designed to run reliably across standard hardware configurations used by MET staff and the general public, including desktop, laptop, and mobile devices. The system shall not depend on specialized or proprietary hardware. Server infrastructure must be capable of supporting concurrent access from both the public dashboard/map and the MET admin panel without performance degradation, and must have sufficient processing capacity to run ML-based flood risk predictions.

2.4.4 Scalability

As the system matures, it is expected to expand beyond the initial set of monitored locations to cover additional areas across Fiji. The system shall be designed so that performance remains consistent as the number of monitored locations, historical weather records, and concurrent users increases. Where practical, the system should scale resources on demand rather than requiring fixed over-provisioning, to keep infrastructure costs manageable within the project's budget constraints

2.4.5 Maintainability

The system should be designed with clear documentation, modular architecture, and version control to ensure future updates such as retraining the prediction model, adding new monitored locations, or extending notification channels (e.g., SMS) can be implemented without requiring significant rework of

existing functionality. Clear separation between the AI/ML component, backend services, and frontend interface is intended to support this long-term maintainability.

2.5 User Documentation

2.5.1 User Manual

A user manual will be provided to assist community users and Fiji MET administrators in using the Flood Risk Prediction and Community Alert System. The manual will contain step-by-step instructions supported by screenshots and explanations of the relevant system functions.

For community users, the documentation will include instructions for registering and logging in, selecting and saving locations, viewing the Flood Risk Dashboard, interpreting flood-risk levels, using the interactive flood-risk map and managing notification preferences.

For Fiji MET administrators, the manual will provide instructions for using administrative functions including managing monitored locations, reviewing predictions, configuring risk thresholds, managing alerts and generating reports.

The user manual will be provided in PDF format so that it can be accessed independently of the web application.

2.6 Assumptions and Dependencies

- ### 2.6.1 Assumptions

- **Weather Data will be Available:** It is assumed that the data regarding the weather, statistics, report levels will all be constantly available by the data providers and meteorological services.
- **Device and Internet Access:** It is assumed that the Public users and Fiji MET staff and administrators will have access to devices like a personal computer or mobile phone that has a supported browser like Google Chrome. It is also assumed that the public users and Fiji MET staff have internet access available.
- **AI Decisions are of Support:** It is assumed that the AI prediction models and this system will act as a support model for Fiji MET staff and not a replacement of the official emergency declarations given by the government.
- **Verification:** It is assumed that the Fiji MET administrators will be able to verify the predictions made by the system before issuing out public alerts in cases of a high risk disaster.

2.6.2 Dependencies

- **External Weather API Data Models:** The system operations depend on 3rd party weather API's to provide necessary data regarding weather status in order for the system to be operational.
- **PostSQL & PostGIS Database:** The system depends on the proper configuration of and operation of the PostSQL database engine along with the PostGIS extensions.
- **Stack & Third Party Libraries:** The backend and prediction features and models all depend on open source software libraries to assist in fast tracking the production of the system. Libraries such as FastAPI and Node.js /React frontend libraries.

- **Browser Third Party Map Render:** The map display is reliant on the third party mapping libraries and the web browser permission settings.

3.0 External Interface Requirements

3.1 User Interfaces

The Flood Risk Prediction and Community Alert System will have a system design that is both user friendly while ensuring that required needs are met. With the use of industry standard styling approaches like Tailwind CSS and CSS Modules, the web application's interfaces will be designed to cater for the needs of the users.

3.1.1 Community/Public Side

- 3.1.1.1 Registration Page

This page is displayed to first time users and is meant for account creation. This form will have required fields that need to be filled out by these users in order to register accounts. These accounts are required to access certain features and functions

- 3.1.1.2 Login Page

Registered users are required to enter their saved credentials for authentication purposes. This is essential for security purposes before giving access to features and functions

- 3.1.1.3 Reset Password Page

In situations where the user cannot remember their passwords, the reset password page allows for the creation of a new password. It works with information saved from the registration page and gives users the ability to reset their password.

- 3.1.1.4 Landing/Home Page

This is the first page the user sees when they have successfully logged in. It will have links to features like the interactive flood risk map and other related functions.

- 3.1.1.5 User Profile

A page that will have all the information regarding the user. This includes information regarding their past site visits, personal details, updating details, change passwords, etc. This page will also display specific information and functions based on the assigned role given by the administrator.

- 3.1.1.6 Flood Risk Dashboard

This dashboard will have risk analysis based on their saved location. Users are also allowed to save multiple locations, these will all be displayed in the dashboard. The dashboard will also have notices and alerts for other weather forecasts.

- 3.1.1.7 Flood Risk Map

This page will have an interactive map of Fiji. This map will display flood risk probabilities on specific monitored locations on the map. Each risk factor percentage will be color coded to help highlight the risk levels.

- 3.1.1.8 Alert Notification View

This page is dedicated for live alerts and will have flood warnings with specific information regarding saved location, risk percentage and necessary actions.

- 3.1.1.9 Location Selection Page

Page with locations across Fiji, Users are able to select and save multiple locations based on preference.

- 3.1.1.10 Support/FAQ

This page is dedicated to answering Frequently Asked Questions . It will also have links to other resourceful tutorials regarding the use of the site. The page will have contact information like email contact and toll free lines.

- 3.1.1.11 Service/About Page

A descriptive page about the site, specifically the functions ,features and other services the site offers.

- 3.1.2 Admin/MET Side

- 3.1.2.1 Login Page

All users with admin roles must log in for authentication and security purposes. This page will determine the access privileges and roles given to the user.

- 3.1.2.2 Admin Dashboard

Contain Analytical information and links to other admin role functions.

- 3.1.2.3 Users Page

This page is dedicated for details regarding user authentication, registration request and other user information. Admin will use this page to give authentication and approve registration requests.

- 3.1.2.4 Alert Management Page

Fiji MET Admin will use this page to manage flood alerts posted on the alert pager. They can choose to keep or delete alerts when needed.

- 3.1.2.5 Threshold Configuration Page

This admin page is used to adjust the configuration for the AI logic. This is especially relevant when adjusting to changing data.

- 3.1.2.6 Monitored Locations Page

Monitored Locations Page - for authorized Fiji MET administrators to manage locations monitored by Flood Risk Prediction and Community Alert System, on the new page.

The page will display the Monitored Locations for the Flood Risk Prediction and Community Alert System. Fiji MET administrators will be able to add a new monitored location, edit a monitored location that is already listed and delete a monitored location if it is no longer required.

3.2 Hardware Interfaces

The Flood Risk Prediction and Community Alert System does not require any specialized hardware interfaces for public users. Public users of the Flood Risk Prediction and Community Alert System will not require any specialized hardware interfaces, as long as they have an Internet connected device that can run a supported web browser (e.g. desktops, laptops, smartphones, etc).

The server infrastructure must provide sufficient processing power, memory, storage and network connectivity to support:

- The PostgreSQL/PostGIS database
- FastAPI backend services
- AI flood-risk prediction processing
- flood-risk map requests
- weather-data processing
- alert processing
- simultaneous user requests

3.3 Software Interfaces

For the Flood Risk Prediction and Community Alert System information will be stored in a relational database management system, i.e. PostgreSQL for the main part of information for the system.

- 3.3.1 Database (PostgreSQL/PostGIS)

PostgreSQL will store structured system information including:

- monitored locations;
- weather and environmental data;
- flood-risk predictions;
- alert information;
- user/account information;
- configured risk thresholds; and
- relevant system records and logs.

The use of the PostGIS extension will allow the use of geographic functionality as required. The data will be stored within the PostgreSQL database using the PostGIS extension to allow the use of coordinates and other spatial data for the monitored locations and the flood-risk map.

The FastAPI backend will use the PostgreSQL/PostGIS database to add, read, update and delete information in the database to support the application.

Access to the database will be restricted to the authorized backend services of the FastAPI application and not granted to public users of the system.

This new rule will mean that historical weather and prediction records will not be able to be modified by users. This rule is consistent with the current rule for the administrators (i.e. they can change risk thresholds but not historical prediction or weather records).

- 3.3.2 ML Model Interface

The system will be connected to an internal machine learning model that will predict flood risks based on the weather data. The machine learning model will get weather data by the FastAPI backend which will send the weather measurements like rainfall, river levels, and locations. Then the model will process these sets of data and in return give a prediction of the flood risk level and a percentage. And if there are data sets missing, then the system identifies the error, records it and goes back to default settings to ensure the system is running.

- 3.3.3 Weather Data API/Source

The system will be connected to an external weather service that automatically collects realtime weather measurements and that weather service is FastAPI. There will be an automatic process that constantly fetches and requests weather data sets from this FastAPI to ensure that the machine learning model has enough and as accurate data as possible to make the correct predictions. Once the data sets are received, the system checks for errors and after that it is stored in the database of PostgreSQL so that it is available to the machine learning model.

- 3.3.4 Notification Service (push)

The system will communicate alert information to users through an internal notification service. Active and new alerts will be pushed to users who have given consent to receive web-based notifications. Using WebSockets, it will allow for the server to push data without a prompt (Diaconu, 2026). Along with a REST polling endpoint available as a fallback for users that have connectivity issues. This is to ensure users are able to receive immediate and reliable notifications.

- 3.3.5 Frontend Framework Interface

The main front end interface will be coded using React, This was chosen as React provides efficient component rendering, scalable modules and other modular components (Olcen & Sokolowska, 2025). This reliability is essential in the overall creation of the systems front end framework. The front end will use styling frameworks like Tailwind CSS. Tailwind CSS is beneficial as it is highly customizable, provides common utility patterns and allows for the system to have complex responsive designs (Team Kranio, 2024). By deploying the framework through the required installation process, the system will be robust, unique and user friendly.

- 3.4 Communication Interfaces

- 3.4.1 Web Protocols

The communication between the web browser and the backend services will use a secure HTTPS with TLS encryption to ensure that the data is secure and protected when it is transferred in transit. The standard requests and answers are done through the JSON through REST endpoints. To ensure that users get real time emergency alerts without having to refresh the page, WebSockets is used to push those alerts instantly onto the web browser as if it is an emergency.

4.0 System Features

4.1 User Registration and Authentication

- 4.1.1 User Account Registration

4.1.1.1 Description

This feature allows new users like Fiji public and Fiji MET staff to register and create new accounts so that they can use and access this system. It is a high priority system feature as an account is required and necessary to be able to use the system, without an account, you cannot use the system.

4.1.1.2 Response Sequences

User Action: The user goes to the account creation page, fills out the necessary forms like name, email address, password and account type, either of the public or MET staff, then presses create account to finalise the credentials.

System Response: The system validates all inputs given by the user, and it uses hashing to securely store the password and records all relevant inputs by the user in the database and waits upon the accounts approval.

4.1.1.3 Functional Requirements.

REQ-1 : The system shall require the user to input a valid email address and a valid password that meets the standard of security protocols.
REQ-2: The system shall not allow duplicate account creations using the same email if it is attempted.
REQ-3: The system shall record all user data and record it in PostgreSQL database.

4.1.1.4 Error Conditions and Responses

Error Condition	Error Response
Empty fields and the user tries to submit the form	The system will force the user to input the missing fields in order to submit the form
Attempt to create an already existing account	Submission Disallowed, display message “Account already exists”

- 4.1.2 User Account Approval (MET admin verifies staff/community accounts)

4.1.2.1 Description

This feature gives the Fiji MET administrators the ability to approve and monitor new account creations by the public and also new Fiji MET staff. This improves security to ensure proper access control and security.

4.1.2.2 Response Sequences

Fiji MET Admin: The admin logs into the system and see's the new accounts and either selects approve or reject on the account creation based on the information provided.

System Response: The system updates the database and enables the new user's access to use the system.

4.1.2.3 Functional Requirements.

REQ-4: The system shall allow the Fiji MET Admin to view the list of all users and new account requests
REQ-5: The system shall update the user account's status based on the admins selection either accepted or rejected
REQ-6: The system shall not allow new users access to the system until it is verified and approved by the administrator.

4.2.1.4 Error Conditions and Responses

Error Condition	Error Response
Account creation request not yet approved	User access not allowed

- 4.1.3 User Login

4.1.3.1 Description

This feature allows public users and Fiji MET staff to log into the system to be able to use it and this is of high priority because in order to use the system, the users must be able to log in to the system.

4.1.3.2 Response Sequences

User/Staff Action: The user or the staff enter their credentials and password in the given fields and click log in.

System Response: The system checks the credentials with the credentials stored in the database, and if the credentials and password matches, and if the account is active and approved by the admin, it grants the user or staff access to the system.

4.1.3.3 Functional Requirements.

REQ-7: The system shall check and verify the email and password entered by the user with the stored credentials in the database.
REQ-8: The system shall send an authentication token to the user when the log in is successful.
REQ-9: Upon successful log in, the system shall direct the user to their appropriate dashboard and page, a public user to the public user's dashboard and the staff to the staff's dashboard.

4.1.3.4 Error Conditions and Responses

Error Condition	Error Response
Invalid Credentials	Display message:" Invalid Credentials", log error
Account yet to be Approved	Display message "Account Approval pending", log error

4.2 Weather Data Management -High Priority

4.2.1 Ingest Weather Data

4.2.1.1 Description

This feature allows the system to automatically retrieve weather and environmental data (such as rainfall, river levels and other relevant measurements) from external weather data sources on a scheduled basis. This ingested data serves as the primary input for the AI flood risk prediction model.

4.2.1.2 Response Sequences

1. The system triggers a scheduled data retrieval job at a defined interval.
2. The system sends a request to the external weather data source/API for each monitored location.
3. The external source returns weather data for the requested location and time period.
4. The system receives the response and passes it the validation process (See 4.2.2).
5. If the retrieval fails, the system logs the failure and retries according to a defined retry policy.

4.2.1.3 Functional Requirements.

REQ 10: The system shall automatically retrieve weather data for each monitored location at a scheduled interval.
REQ 11: The system shall support retrieval from at least one external weather data API/source.
REQ 12: The system shall log the timestamp and outcome (suceess/failure) of every data retrieval attempt.
REQ 13: The system shall allow an admin to manually trigger a data retrieval for a specific location outside the scheduled event.

4.2.1.4 Error Conditions and Responses

Error Condition	Error Response
External weather source unavailable	Log the failure, retry per policy and flag the location as “data unavailable"if retries are unavailable.
Malformed or unexpected response format	Log the error and discard the response without passing it to validation.
Network/connection timeout	Retry per policy; log the timeout
No data available for the requested location/time	Log as “no data returned”and proceed without blocking other locations.

- 4.2.2 Validate & Store Weather Data

4.2.1.1 Description

This feature validates incoming weather data for completeness and plausibility before storing it in the database, ensuring that flood risk predictions are not generated from corrupted, incomplete, or clear data errors.

4.2.1.2 Response Sequences

1. The system receives raw weather data from the ingestion process(4.2.1).
2. The system checks the data against expected fields, types, and plausible value ranges (e.g. rainfall cannot be negative).

3. If the data passes validation, the system stores it in the database, linked to the relevant location and timestamp.
4. If the data fails validation, the system discards or flags the record and logs the reason.
5. The system makes newly stored, validated data available to the AI prediction model

4.2.1.3 Functional Requirements.

REQ 14: The system shall validate all required weather data fields before storing a record.
REQ 15: The system shall validate that weather data values fall within plausible ranges (e.g. non-negative rainfall, realistic temperature bounds) before storing.
REQ 16: The system shall reject and log any weather data record that fails validation, without storing it.
REQ 17: The system shall make newly validated weather data available to the AI prediction module without manual intervention.
REQ 18: The system shall prevent duplicate weather data records for the same location and timestamp,

4.2.1.4 Error Conditions and Responses

Error Condition	Error Response
Missing required field	Reject the record and log which field(s) were missing.
Value outside plausible range	Reject the record and log the invalid value and expected range
Duplicate record for same location/timestamp	Discard the duplicate and log the occurence
Database or server unavailable during storage	Cache the validated data locally and retry storage once service is restored

4.3 AI Flood Risk Prediction -High Priority

- 4.3.1 Generate Risk Prediction

4.3.1.1 Description

This feature processes historical data using AI models. The processed data is then used to generate the localized flood risk predictions. These predictions are then distributed to risk levels, each is associated with its urgency **(** (LOW/MODERATE/HIGH/VERY HIGH). The risks are updated in real time.

4.3.1.2 Response Sequences

1. The system receives weather and environmental data from Fiji MET and other external sources.
2. The Prediction Model processes data and calculates flood risk levels based on requested location.
3. Store predictions made.
4. Send Risk level to frontend visuals (interactive map)

4.3.1.3 Functional Requirement

REQ 19: The system shall process any received past and present meteorological data.
REQ 20: The system shall create flood risk predictions based on the processed data.
REQ 21: The System shall organise and categorize risk predictions by urgency levels.
REQ 22: The system shall update predictions every 30 mins or when new data is made available

4.3.1.4 Error Conditions and Responses

Error Condition	Error Response
Data Source Unavailable	Used previously saved data and update prediction when current data is available.
Execution failure of AI model	Retry prediction, log error for review.
Database or server unavailable during	Ques all requests until the service runs again.

- 4.3.2 Display Risk by Location

4.3.2.1 Description

This feature allows for the system to display predicted flood risk by location. It will also display any other relevant meteorological information. The risk levels are displayed on an interactive map allowing for users to quickly view their requested location.

4.3.2.2 Response Sequences

1. The user selects a location via interactive map or search.
2. The system retrieves and displays flood risk predictions based on requested location.
3. The system displays flood risk levels with the associated color-coded indicator.
4. Users have access to additional meteorological information.

4.33.2.3 Functional Requirement

REQ 20: The system shall allow users to select location by interactive map or via search bar
REQ 21: The system shall display flood risk for the requested location.
REQ 22: The system shall display past meteorological information for the requested location.

4.3.2.4 Error Conditions and Responses

Error Condition	Error Response
Data for the selected location not found.	Used previously saved data and update prediction when current data is available.
Current Predictions Unavailable	Display previous risk predictions with timestamp
Rendering Error	Reload map; alt text displayed.

4.4 Alert Management - High Priority

- 4.4.1 Create/Approve Alert

4.4.1.1 Description

This feature allows the system to automatically generate a candidate alert when the AI model predicts an elevated flood risk (High or Very High) for a monitored location. The candidate alert is placed in a pending state and must be reviewed and approved by MET admin before it becomes visible to the public. Admins also have the option to manually create an alert independent of a model prediction, if the situation calls for an external alert.

4.4.1.2 Response Sequences

1. The AI model generates a risk prediction for monitored location.
2. If the predicted risk level is High or Very High, the system automatically creates a pending alert for a monitored location and prediction.
3. The admin navigates the ‘Alert Management' page and views the appending alert in the list.
4. The admin reviews the alert details, including risk level, confidence and affected area(s).
5. The admin approves and/or edits the alert message and severity before sending.
6. The system changes the alert status to "approved", records the approving admin, timestamp and pushes the alert to all connected clients.
7. Alternatively, the admin selects “Create Alert” manually, fills in location, severity and message, and submits it directly for approval.

4.4.1.3 Functional Requirements

REQ 23: The system shall automatically create a pending alert when a prediction returns a risk level of High or Very High for a monitored location.
REQ 24: The system shall allow an admin to manually create a new alert for a location, independent of a model-generated prediction.
REQ 25: The system shall display all pending alerts to admins, including location, risk-level, confidence score, and prediction timestamp.
REQ 26: The system shall allow the admin to edit an alert’s message or severity before approving it.
REQ 27: The system shall record the identity of the approving admin and the approval timestamp for every approved alert

4.4.1.4 Error Conditions and Responses

Error Condition	Error Response
Required fields are blank (manual alert creation)	Display error message highlighting missing fields
Invalid location selected	Display error messages stating the location does not exist
Database or server unavailable during approval	Cache the alert location locally and display “Service temporary unavailable, action will be retried”
Duplicate pending alert for same location or prediction	Display existing pending alert instead of creating a duplicate

- 4.4.2 Override/Suppress Alert

4.4.2.1 Description

This feature allows a MET admin to override or suppress an alert that has already been created, in cases such as a false positive, outdated prediction or new information that changes the assessment.

4.4.2.2 Response Sequences

1. The admin navigates to the “Alert Management” page and selects an active or pending alert.
2. The admin selects “Override” to change the severity level, or "Suppress” to prevent the alert from being shown publicly.
3. The system prompts the admin to confirm the action and for suppression, provide a reason.
4. The system updates the alert’s status and severity accordingly.
5. If the alert was already public, the system immediately updates or removes it from all connected clients.
6. The system logs the override/suppression action, the admin who performed it, the reason (when applicable) and the timestamp

4.4.2.3 Functional Requirements

REQ 28: The system shall allow an admin to override the severity level of an existing alert.
REQ 29: The system shall allow an admin to suppress an alert, removing or preventing it from public display.
REQ 30: The system shall require a reason to be entered when suppressing an alert.
REQ 31: The system shall immediately update all connected clients when an already-public alert is overridden or suppressed.
REQ 32: The system shall record the log of every override or suppression action with the responsible admin, reason, and timestamp
REQ 33: The system shall prevent a suppressed alert from being re-approved without an explicit reset action by an admin

4.4.1.4 Error Conditions and Responses

Error Condition	Error Response
Missing suppression reason	Display error message requiring suppression reason
Alert no longer exists (Already resolved)	Display error messages stating "The alert is no longer active"
Database or server unavailable	Cache the alert location locally and display "Service temporary unavailable, action will be retried"

- 4.4.3 Configure Risk Thresholds

4.4.3.1 Description

This feature allows the system to view and adjust the risk score threshold that determines which severity tier (Low, Moderate, High, Very High) a prediction falls into and which tiers automatically trigger a pending alert.

4.4.3.2 Response Sequences

1. The admin navigates to the "Threshold Configuration"page.
2. The system displays the current threshold values for each severity tier.
3. The admin edits one or more threshold values.
4. The system submits the change.
5. The system validates the threshold remains in a logical ascending order and saves the update.
6. The system applies the new thresholds to all future predictions, past alerts are not changed.

4.4.3.3 Functional Requirements

REQ 35: The system shall display the current risk score threshold for each severity tier to admins
REQ 36: The system shall allow an admin to update the threshold values for each severity tier.
REQ 37: The system shall validate that threshold values in ascending order before saving.
REQ 38: The system shall apply updated thresholds only to predictions generated after the change but not prior to the change.
REQ 39: The system shall record the log threshold changes with the responsible admin and timestamp.

4.4.3.4 Error Conditions and Responses

Error Condition	Error Response
Threshold entered out of ascending order	Display error message highlighting threshold must increase by severity
Invalid value entered e.g. non numeric key or missing threshold value	Display field validation error message
Database or server unavailable during approval	Display “Service temporary unavailable”

4.5 Alert Notification -High Priority

- 4.5.1 Display Public Alert

4.5.1.1 Description

This feature displays approved, active alerts to the public on the dashboard and flood risk map, without requiring the user log in.

4.5.1.2 Response Sequences

1. The user opens the dashboard or map page.
2. The system requests the current list of active alerts.
3. The system displays each active alert with its severity, affected location, and a short advisory message, using colour coding matching the severity of each alert (Red, Orange, Yellow and Green)
4. If a new alert is approved while the user has the page open, the system pushes the update via WebSocket and displays it without requiring a page refresh.
5. When an alert is resolved or suppressed by an admin, the system removes or updates it on the user’s view in real time.

4.5.1.3 Functional Requirements

REQ 40: The system shall display all active, approved alerts on the public dashboard and map without requiring authentication.
REQ 41: The system shall visually distinguish alert severity using consistent colour across the dashboard and map.
REQ 42: The system shall update a connected client’s view of active alerts within 1 second of an alert being approved, overridden or suppressed.
REQ 43: The system shall provide a REST endpoint to fetch the current list of active alerts for clients that cannot maintain a WebSocket connection.
REQ 44: The system shall display the affected location, severity and advisory message active alert.

4.5.1.4 Error Conditions and Responses

Error Condition	Error Response
WebSocket connection fails or drops	Automatically fall back to REST polling until the connection is restored
No active alerts for the requested location	Display “ No active alerts for this location”
Database or server unavailable	Display “Unable to load alert information, please try again”

- 4.5.2 Push Notification

4.5.2.1 Description

This feature allows users who have opted to receive notifications through an optional browser push channel. When a new alert is approved for a location they follow, in addition to the alert being visible on the dashboard.

4.5.2.2 Response Sequences

1. A registered user opts in to notifications to follow.
2. When an alert is approved for the following location, the system queues a notification for that user.
3. The user receives the notification through the notification log.
4. The user receives the notification and if they wish, they can click on the notification for a more detailed view.

4.5.2.3 Functional Requirements

REQ 45: The system shall allow a registered user to opt in to push notifications for one or more followed locations.

REQ 46: The system shall send a notification to opted-users within 2 minutes of alert being approved for a followed location.

REQ 47: The system shall allow a user to opt out of notifications at any time.

REQ 48: The system shall not send a notification for an alert that is later suppressed before the delivery is completed.

4.5.2.4 Error Conditions and Responses

Error Condition	Error Response
Web Push fails to send	Retry sending and log the error
Push notification permission not granted by browser	Fall back silently to dashboard-only display; do not block other functionality
Database or server unavailable	Queue the notification and retry once service is restored

- 4.6 Flood Risk Map -Medium Priority

- 4.6.1 View Map with Risk Overlay

4.6.1.1 Description

This feature gives users the ability to view flood risks on an interactive map. The risk levels will be overlaid on the map and will display different risk levels using colors (for e.g green= low risk, red = high risk). This map will be updated in real time based on the predictions made by the AI model. Fiji MET staff and the general public will be able to determine the areas at most risk of flooding

4.6.1.2 Response Sequences

1. User Navigates to the user dashboard and access the map view.
2. The user reads real-time flood risk data displayed on map from the AI model.
3. The User reads Map which displays risk levels (LOW/MODERATE/HIGH/VERY HIGH) on specific locations on the map.

4. The User can interact with the map for additional information, zooming to specific locations, etc.

4.6.1.3 Functional Requirements

REQ 49: The system shall display an interactive map with color-coded flood risk levels
REQ 50: The system shall update flood risk overlays in real time when the AI model makes new predictions.
REQ 51: The system shall allow interactive features like zooming and panning on the map for detailed location views.
REQ 52: The system shall allow selecting a specific location with predicted information.

4.6.1.4 Error Conditions and Responses

Error Condition	Error Response
Map loading Failures	Fallback message displayed “Map unavailable please try again”
Map Data Fails to load	Show last recorded risk data with timestamp
Network Connectivity Issues	Message displayed “network failure, retry loading map”

- 4.6.2 View Location Detail

4.6.2.1 Description

This feature allows for users to select specific locations for meteorological information, this also applies for flood risk data. Arthritis can use this to monitor and identify flood risks for multiple locations.

4.6.2.2 Response Sequences

1. Users select a location on the map or via search bar.
2. The system displays the selected location including flood risk data.
3. The system displays past data, current risk level and upcoming alerts.

4.6.2.3 Functional Requirements

REQ 53: The system shall allow users to select specific locations via the interactive map or search bar.
REQ 54: The system shall display current flood risk levels for the specific selected location.
REQ 55: The system shall display past data for the selected location.
REQ 56: The system shall display alerts associated with the selected location.

4.6.2.4 Error Conditions and Responses

Error Condition	Error Response
Location data not found	Fallback message displayed “Location data unavailable”
Database Query Fail	Retry query request,log error and show request
Server Unavailable	Queue user request and notify when information requested is available.

- 4.7 Monitored Location Management -Medium Priority

- 4.7.1 Add/Edit/Remove Monitored Location

- 4.7.1.1 Description

Authorized MET admin users are able to add,edit and delete locations that the system is monitoring. The locations that are being monitored by the system are used to provide location based weather, flood risks and alerts.

4.7.1.2 Response Sequences

1. The MET admin accesses the Monitored Locations page.
2. The system displays the list of currently monitored locations.
3. The admin chooses to add a new location, edit an existing monitored location or remove a monitored location.
4. For adding or editing, the admin enters or updates the required location information.
5. The system validates the location information.
6. For removing a location from the Monitored Locations, the admin is asked to confirm the removal of the selected location.
7. The system updates the monitored location information in the database.
8. The system displays the updated list of monitored locations.

4.7.1.3 Functional Requirements

REQ 57: The system shall allow an authorized MET admin to add a new monitored location.
REQ 58: The system shall allow an authorized MET admin to edit an existing monitored location.
REQ 59: The system shall allow an authorized MET admin to remove an existing monitored location.
REQ 60: The system shall validate the required location before adding or updating a monitored location.
REQ 61: The system shall request confirmation from the MET admin before removing a monitored location.
REQ 62: The system shall store changes to monitored locations in the PostgreSQL/PostGIS database.
REQ 63: The system shall restrict monitored location management to users with MET admin privileges.

4.7.1.4 Error Conditions and Responses

Error Condition	Error Response
Required location information is missing	Display an error message and request the admin to complete the required fields
Invalid location information is entered	Display an error message and request valid location information
Location removal fails	Keep the location in the monitored locations list and inform the admin that the removal was unsuccessful

- 4.8 Reporting & Analytics -Medium Priority

- 4.8.1 View Key Metrics

- 4.8.1.1 Description

MET admin users can view system metrics such as flood risk prediction, monitored locations and alerts. Authorized users can use these metrics to view current flood risk and system activity in a quick view format.

4.8.1.2 Response Sequences

1. The MET admin accesses the reports page.
2. The system retrieves the latest available data from the database.
3. The system displays any key metrics related to monitored locations, flood risk predictions and alerts.
4. The admin reviews the displayed metrics.
5. The admin can select available information to view further details where applicable.

4.8.1.3 Functional Requirements

REQ 64: The system shall allow an authorized MET admin to view key system metrics.
REQ 65: The system shall display information relating to monitored locations.
REQ 66: The system shall display available information relating to current flood risk predictions.
REQ 67: The system shall display relevant alert information available within the system.
REQ 68: The system shall retrieve reporting information from the stored system data.
REQ 69: The system shall restrict access to reporting and analytics functions to authorized MET admin users.

4.7.1.4 Error Conditions and Responses

Error Condition	Error Response
Database or server unavailable	Display a service unavailable message and allow the admin to retry later
Metrics fail to load	Display an error message and provide the option to retry
User does not have MET admin privileges	Deny access to the reporting and analytics functions

- 4.8.2 Generate Report

- 4.8.2.1 Description

Reports are created by authorized MET admin users and can contain any information within the system. This allows MET admin users to see what has happened previously on the system and associated flood risk information.

4.8.2.2 Response Sequences

1. The MET admin accesses the reports page.
2. The admin selects the option to generate a report.
3. The system displays the available report criteria.
4. The admin selects the required report information and data range.
5. The system validates the selected criteria.
6. The system retrieves the information from the database.
7. The system generates and displays the report.
8. The admin reviews the generated report.

4.8.2.3 Functional Requirements

REQ 70: The system shall allow an authorized MET admin to generate reports using stored system information.
REQ 71: The system shall allow the admin to select a date range for the report.
REQ 72: The system shall allow reports to include relevant flood risk prediction information.
REQ 73: The system shall allow reports to include relevant monitored location information.
REQ 74: The system shall allow reports to include relevant alert information.
REQ 75: The system shall display the generated report to the MET admin.
REQ 7: The system shall notify the admin when no information matches the selected report criteria.
REQ 8: The system shall restrict report generation to authorized MET admin users.

4.8.2.4 Error Conditions and Responses

Error Condition	Error Response
Invalid date range is entered	Display an error message and request a valid date range
No records match the selected criteria	Display a message informing the admin that no information was found
Report generation fails	Display an error message and allow the admin to try again

5.0 Other Non-Functional Requirements

- 5.1 Performance Requirements

The system is required to receive, retrieve and send vital weather information from the database during expected and unexpected environments.

NF REQ 1: The mean response time for the system shall be less than 3 seconds under normal operating conditions and 6 seconds during abnormal conditions.

NF REQ 2: The system shall generate a flood risk prediction within 5 seconds of a request being submitted.

NF REQ 3: The system shall be able to recover from critical failure within 1 hour during normal operating conditions without data loss or corruption

NF REQ 4: The system shall send out a new or updated alert to all connected users within 1 second of the alert being issued or approved.

NF REQ 5: The system shall be available 99% of the normal operating conditions, regardless of the system's role in time-sensitive flood warnings.

NF REQ 6: users who have opted for web-based push notifications shall receive alert notifications within 2 minutes of alerts being approved.

NF REQ 7: The system shall support at least 150 users accessing the dashboard and map concurrently without data inconsistency or performance degradation.

- 5.2 Safety Requirements

The system shall be designed to ensure integrity of weather data, predictions and alerts. Furthermore, to prevent unauthorized or accidental changes from causing incorrect or missing warnings from being sent out.

NF REQ 8: The system shall only allow MET admin users to approve, update or revoke a weather warning.

NF REQ 9: The system shall log every alert action (created, approved, updated or revoked) with a timestamp and corresponding traceback of the user(s) associated with the log.

NF REQ 10: If the ML model is unavailable or fails to return a prediction, the system shall display the most recently valid prediction for that location along with a timestamp.

NF REQ 11: If the weather data feed fails or returns incomplete data, the system shall flag the affected location as " "data unavailable instead of returning faulty predictions.

NF REQ 12: In the event of a failure during alert creation, the alert shall be cached so that it can be reattempted once the system has recovered.

NF REQ 13: Since the system stores user and location data. Sensitive data transmission and storage shall be encrypted to prevent data theft/leaks.

NF REQ 14: The system shall enforce strong password creation policies for all admin and registered accounts.

- 5.3 Security Requirements

Security requirements are designed to protect the system from unauthorized access that could be used to misuse alert functionality, change flood risk data, or compromise user privacy.

NF REQ 15: The system shall require users to authenticate before accessing any admin features, including alert flood risk data or compromise user privacy.

NF REQ 16: All sensitive data, including user account details, shall be encrypted in transit (HTTPS) and encrypted or hashed at rest.

NF REQ 17: The system shall ensure secure handling of user and location data in accordance with applicable Fiji data protection requirements from Fiji CERT.

NF REQ 18: The system shall provide a secure password reset mechanism for registered users.

NF REQ 19: The system shall restrict alert threshold configuration

- 5.4 Usability Requirements

NF REQ 20: The system shall use responsive design so the dashboard, map and alert views remain usable across desktop and mobile devices.

NF REQ 21: The system shall maintain a consistent layout, navigation and terminology across all pages to ensure an ergonomics throughput site.

NF REQ 22: Flood risk information shall be preserved in plain, non-technical language and clear visual indicators (e.g. Colour -coded severity) so it is understandable to the general public and those without a technical background.

NF REQ 23: The system shall provide onboarding guidance (e.g. a short walkthrough or FAQ) to help first time users understand how to view and interpret alerts and risk information.

- 5.5 Business Rules

BR 1: Only users with MET admin privileges may access the admin panel and its related functionality (alert management, threshold configuration, local management).

BR 2: Community users are not required to authenticate to view public flood risk information or active alerts, but must authenticate to manage personal alert notification preferences.

BR 3: All new admin accounts must be approved by an existing MET admin before gaining access to the system

BR 4: The AI model may generate a predicted risk level, but only a MET admin may approve, override, or suppress an alert before it reaches the public.

BR 5: Admins may adjust risk thresholds but may not alter historical prediction or weather records.

BR 6: Every alert action must be attributed to a specific admin account; unlogged alert actions are not permitted.

6.0 Other Requirements

The Flood Risk Prediction and Community Alert System is developed for Fiji Metrological Services and deployed within Fiji. The system's timestamps and alert scheduling shall be based on Fiji Standard Time (GMT 12+). The system shall be designed to comply with Fiji CERT (Fiji CERT, 2026) data privacy requirements, as well as any relevant Fiji MET terms of service or data-sharing agreements outlining the use of weather and location data.

7.0 Appendix

Glossary

Fiji MET	Fiji Metrological Services
ML	Machine Learning
SRS	Software Requirements System
API	Application Programming Interface that allows the system to communicate with external systems that will improve system functionality.
Flood Risk Category - LOW	Indicates that the predicted risk of flooding for a location is low.
Flood Risk Category - MODERATE	Indicates an increased flood risk that may require monitoring
Flood Risk Category - HIGH	Indicates a significant predicted flood risk where warnings or increased preparedness may be required.
Flood Risk Category - VERY HIGH	Indicates the highest predicted flood-risk level and requires immediate attention and warning.

8.0 Mark Allocation Sheet

Group 14
Project Manager: Parvish Mohan

Student ID	Contribution %
S11230414	70
S11225654	100
S11229870	70
S11220998	100
S11231286	100

Group Member	Student ID	Signature	Date
Parvish Mohan	S11230414	Parvish Mohan	28/08/2026
Aryan Sewak	S11225654	Aryan Sewak	28/08/2026
Rishwant Kumar	S11229870	Rishwant Kumar	28/08/2026
Louvatu Vosailagi	S11220998	Louvatu Vosailagi	28/08/2026
Atharv Narayan	S11231286	Atharv Narayan	28/08/2026